

Model-independent analysis of DESI DR1 full shape + BAO: parameter recovery with the BAO-template model **Ag245F**

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What is being tested

The model-independent (MI) analysis fits the DESI DR1 full-shape and BAO data with the linear power spectrum, the growth rate and the distances at each redshift as free parameters, instead of computing them from a cosmology. A cosmological model θ is then fitted by projecting the MI posterior onto it, with no new run of the EFT likelihood:

$$p(\theta | d) \propto \frac{p_{\text{MI}}(\phi(\theta) | d)}{\pi_{\text{MI}}(\phi(\theta))} \pi(\theta).$$

Test. For four cosmologies we compare this projection with a *direct* fit of the same data, made with the full EFT likelihood and the same priors. The projection passes if every parameter shifts by less than 0.3σ and every width agrees within 20%. Here a shift is $(\bar{x}_{\text{MI}} - \bar{x}_{\text{direct}})/\sigma_{\text{direct}}$ and a width ratio is $\sigma_{\text{MI}}/\sigma_{\text{direct}}$.

Data and theory

DESI DR1 P_0 , P_2 of BGS, LRG1–3, ELG2 and QSO ($z_{\text{eff}} = 0.295\text{--}1.491$), $0.02 < k < 0.20 h \text{ Mpc}^{-1}$, with the survey window. Added to them are the 10 post-reconstruction BAO dilations, in one joint covariance: 442 data points. The theory is the pybird one-loop EFT (west-coast basis), IR-resummed and with AP, evaluated through the 80-knot loop emulator.

The MI model

$$P_{\text{lin}}(k, z_i) = \left[\frac{D(z_i)}{D(z_N)} \right]^2 a(k) T_{\text{nw}}(k) \left[1 + (1 + A) e^{-k^2 \Delta \Sigma^2 / 2} O(\alpha_{r_s} k) \right], \quad z_N = 1.491.$$

- T_{nw} and O : the no-wiggle part and the wiggle pattern of the fiducial (Planck 2018) linear spectrum.
- $a(k)$: a cubic spline in $\ln k$ on 21 nodes (spacing 0.44). That is too coarse to move a wiggle, so the BAO position comes only from α_{r_s} and the dilations.
- α_{r_s} , A , $\Delta \Sigma^2$: the sound-horizon rescaling, amplitude and damping of the wiggles, as in a BAO template fit.
- Per redshift: $f(z_i)$, $D(z_i)/D(z_N)$, and $\alpha_{\parallel}$, $\alpha_{\perp}$ for the full-shape AP. The BAO points are predicted as α/α_{r_s} , so there is no separate BAO dilation.
- b_1 , c_2 per sample are sampled; the other 42 EFT coefficients are profiled analytically.

That makes 59 sampled parameters. The priors are $\ln \alpha_{r_s} \sim \mathcal{N}(0, 0.05)$, $A \sim \mathcal{N}(0, 0.3)$, $\Delta\Sigma^2 \sim \mathcal{N}(0, 25 \text{ Mpc}^2)$, and $\ln f, \ln \alpha_{\parallel, \perp} \sim \mathcal{N}(\text{fid}, 0.3)$. The $\ln a$ shape has $\sigma = 0.5$ per node plus a smoothness penalty, and the per-redshift amplitudes have $\sigma(\ln P) = 0.6$. The full list is in `WIGGLE_SETUP.md`, in the repository folder `notebooks/desi_fsbao`.

MI chain: NUTS, 32 chains $\times$ 2000 draws, max $\hat{R} = 1.005$, min ESS 6405. Best-fit $\chi^2 = 284.2$ for 442 points; the direct Λ CDM best fit has 332.3.

Projection and reference

- $\phi(\theta)$: for each θ it holds that cosmology's f , D , $\alpha_{\parallel}$, $\alpha_{\perp}$ and α_{r_s} .
- $a(k)$, A , $\Delta\Sigma^2$: they come from a least-squares fit of θ 's linear spectrum on the 80 emulator knots, weighted by the data's Fisher information. This fit uses θ only, never the data, so the BAO amplitude and damping are *predicted* by the cosmology.
- p_{MI} : an ensemble of 10 normalizing flows (RealNVP) trained on the MI chain.
- **Reference:** a direct NUTS chain of the *true* likelihood for each cosmology. Its linear spectrum (CosmoPower, or the ede-v2 emulator for EDE) goes to the same EFT, AP and BAO, with the same priors: flat boxes plus $\omega_b \sim \mathcal{N}(0.02218, 0.00055)$ (BBN) wherever ω_b is free.

Result

cosmology	largest shift (flows)	width ratios (flows)	largest shift (Gaussian p_{MI})
ΛCDM ($\omega_{\text{cdm}}, A_s, h$)	0.18σ	0.97–1.03	0.18σ
$\Lambda\text{CDM} + \omega_b$ (BBN), n_s	0.26σ	0.97–1.02	0.25σ
$w_0w_a\text{CDM}$	0.20σ	1.00–1.05	0.66σ
EDE (n_s fixed)	0.11σ	0.96–1.02	0.19σ

All four pass with the flows. The Gaussian estimate misses $w_0w_a\text{CDM}$ along h – w_0 – w_a : the MI posterior is non-Gaussian there, because α_{r_s} and the dilations are nearly degenerate. The flows are therefore the default.

cosmology	parameter	direct (true likelihood)	MI $\rightarrow$ cosmology	shift [σ]	width ratio
ΛCDM	ω_{cdm}	0.1211 ± 0.0048	0.1203 ± 0.0047	-0.18	0.97
	$\ln 10^{10} A_s$	3.024 ± 0.092	3.040 ± 0.090	$+0.18$	0.98
	h	0.6882 ± 0.0061	0.6879 ± 0.0063	-0.06	1.03
$\Lambda\text{CDM}, \omega_b, n_s$	ω_{cdm}	0.1162 ± 0.0063	0.1146 ± 0.0061	-0.26	0.98
	ω_b	0.02206 ± 0.00054	0.02208 ± 0.00053	$+0.02$	0.99
	n_s	1.006 ± 0.040	1.014 ± 0.039	$+0.18$	0.97
	$\ln 10^{10} A_s$	3.079 ± 0.104	3.106 ± 0.103	$+0.26$	0.99
	h	0.6850 ± 0.0074	0.6845 ± 0.0075	-0.08	1.02
$w_0w_a\text{CDM}$	ω_{cdm}	0.1227 ± 0.0079	0.1214 ± 0.0081	-0.17	1.03
	ω_b	0.02209 ± 0.00053	0.02210 ± 0.00054	$+0.01$	1.01
	n_s	0.971 ± 0.045	0.980 ± 0.046	$+0.20$	1.01
	$\ln 10^{10} A_s$	2.852 ± 0.173	2.882 ± 0.172	$+0.17$	1.00
	h	0.667 ± 0.021	0.667 ± 0.022	$+0.01$	1.05
	w_0	-0.64 ± 0.23	-0.66 ± 0.24	-0.08	1.05
	w_a	-1.46 ± 0.84	-1.37 ± 0.85	$+0.10$	1.01
EDE	ω_{cdm}	0.1374 ± 0.0111	0.1365 ± 0.0112	-0.09	1.02
	ω_b	0.02224 ± 0.00054	0.02224 ± 0.00054	-0.00	0.99
	$\ln 10^{10} A_s$	3.034 ± 0.106	3.045 ± 0.102	$+0.11$	0.96
	h	0.730 ± 0.029	0.729 ± 0.030	-0.03	1.01
	f_{EDE}	0.152 ± 0.087	0.148 ± 0.085	-0.05	0.98
	$\log_{10} z_c$	3.54 ± 0.41	3.55 ± 0.41	$+0.03$	1.00
	θ_i	1.55 ± 0.86	1.54 ± 0.85	-0.01	0.99

Why A and $\Delta\Sigma^2$ are predicted, not marginalized

Marginalizing A and $\Delta\Sigma^2$ in the projection treats them as free nuisances of the cosmology fit, which throws away what the wiggle amplitude and damping say about ω_b , ω_m and EDE. With them marginalized, the $\Lambda\text{CDM} + \omega_b, n_s$ widths grow by $\times 1.24$ (ω_{cdm}) and $\times 1.29$ (n_s). EDE shifts by $+0.37$ to $+0.44\sigma$ in ω_{cdm} , h and f_{EDE} (Gaussian estimate). A direct true-likelihood fit with A , $\Delta\Sigma^2$ free shifts the same way, so this is lost information, not a bias of the method.

On their own the data barely constrain them: $A = 0.16 \pm 0.16$ and $\Delta\Sigma^2 = 7 \pm 21 \text{ Mpc}^2$. ΛCDM predicts $A = -0.02 \pm 0.07$ and $\Delta\Sigma^2 = -1.5 \pm 4.4 \text{ Mpc}^2$, consistent with the data. Their correlation with every BAO dilation is $|r| \leq 0.10$, so the BAO position is carried by α/α_{r_s} alone.

Caveats

- r_d : the ω_b exponent of pybird’s analytic r_d fit has the wrong sign ($+0.13$ instead of -0.13). Every chain here uses the corrected fit.

- **$\Delta\Sigma^2$ prior:** at 25 Mpc^2 it is wider than physical cosmologies need. A $\sim 10 \text{ Mpc}^2$ prior, or setting O to zero above the data range, would stop its negative tail from amplifying the wiggles at $k \gtrsim 0.3 \text{ Mpc}^{-1}$.
- **Absolute amplitude:** the absolute linear amplitude is degenerate with b_1 (Fig. 5); the *shape* of P_{lin} is what the data constrain.

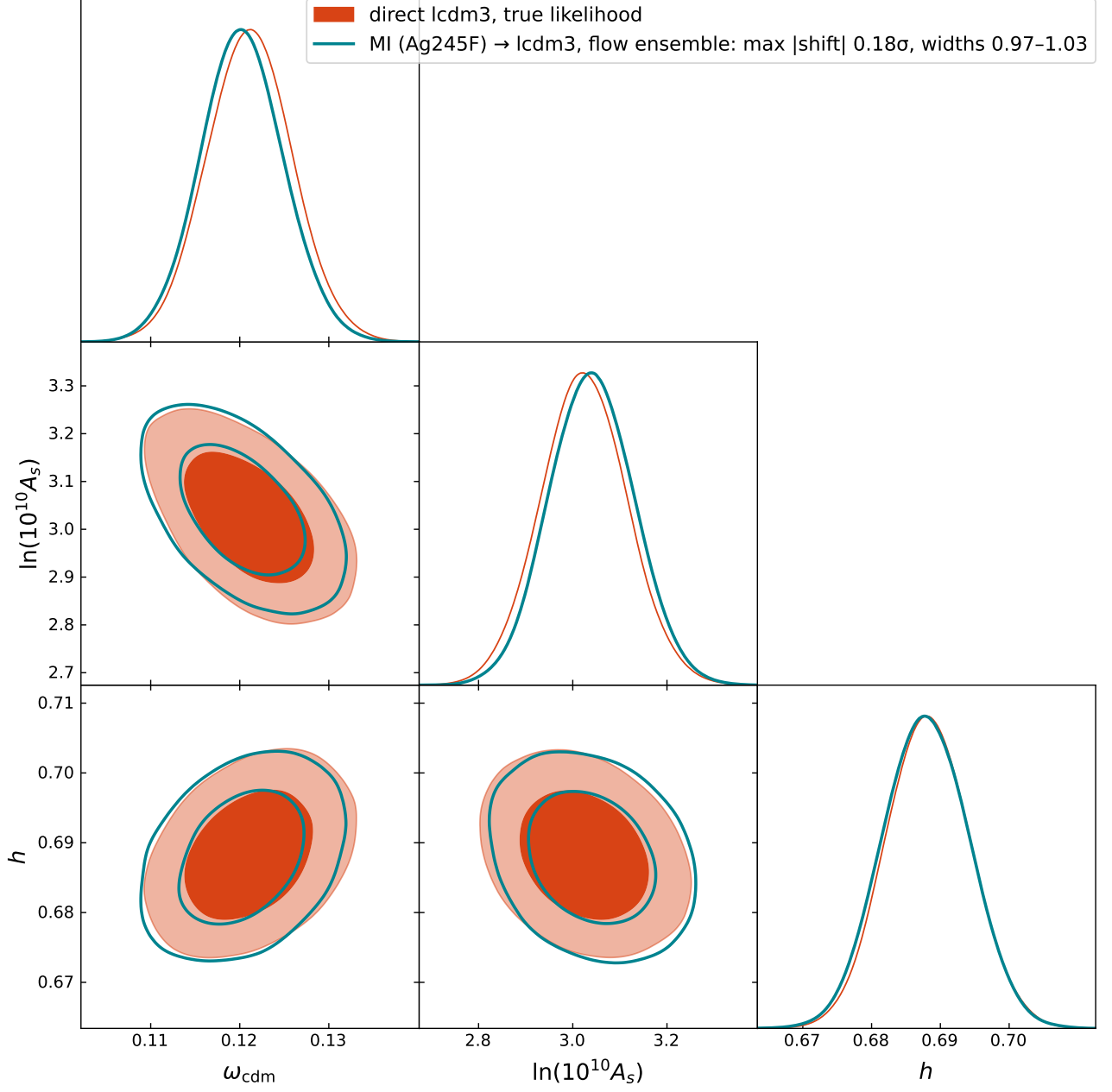


Figure 1: Λ CDM (ω_{cdm} , $\ln 10^{10} A_s$, h): direct true-likelihood posterior (filled) against the MI posterior projected onto Λ CDM (flow ensemble).

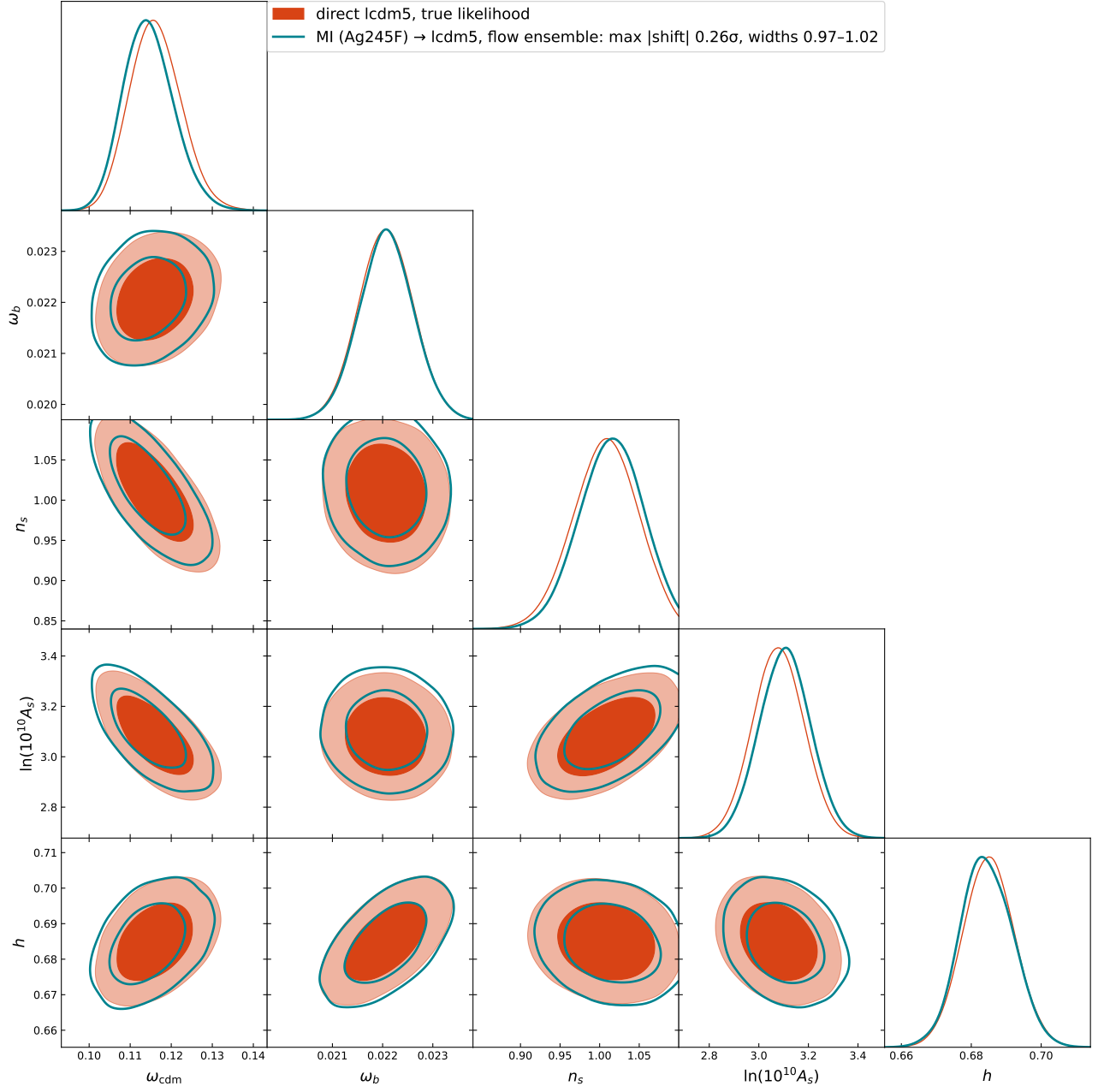


Figure 2: Λ CDM with ω_b (BBN prior) and n_s free.

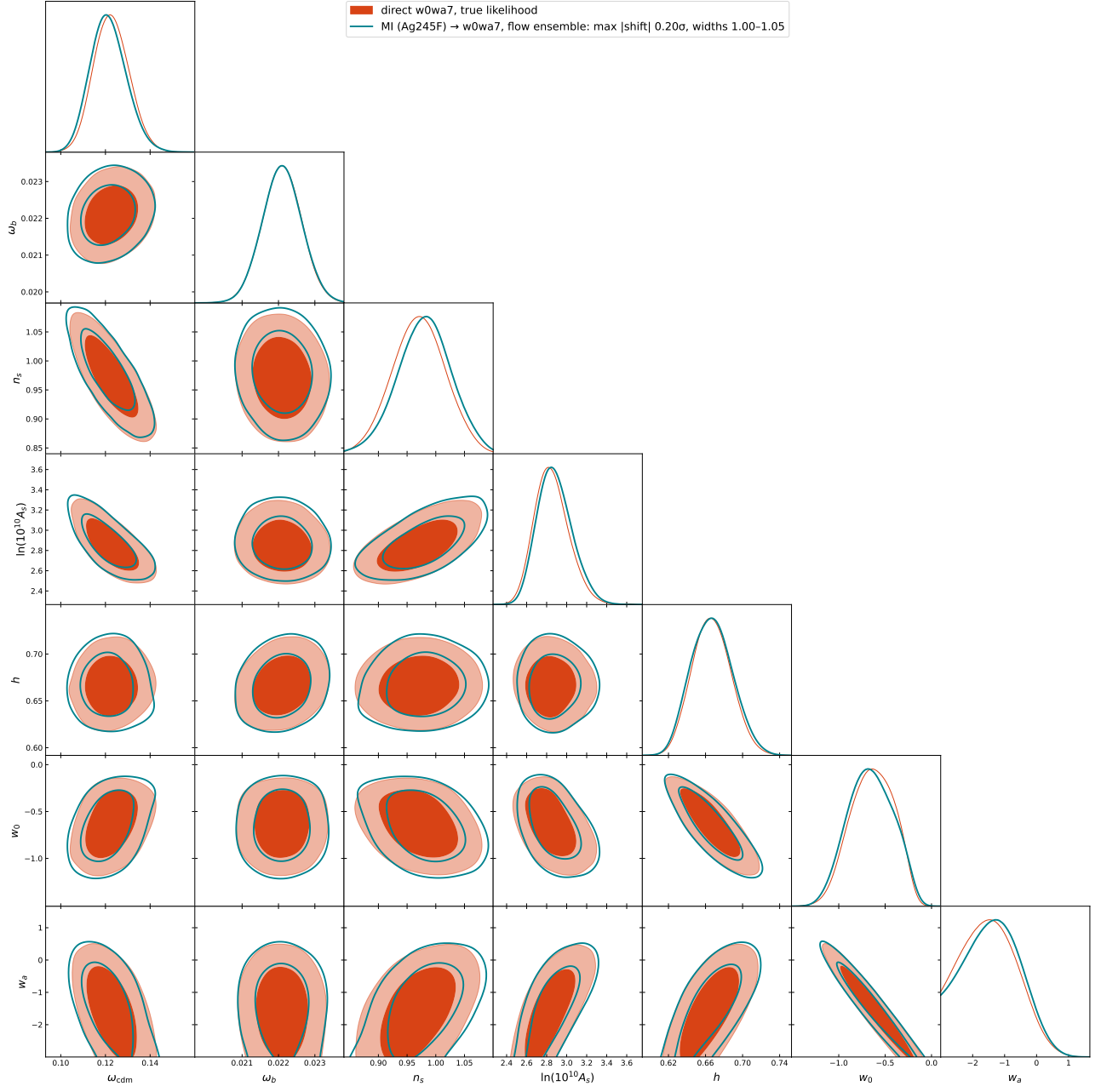


Figure 3: $w_0 w_a$ CDM.

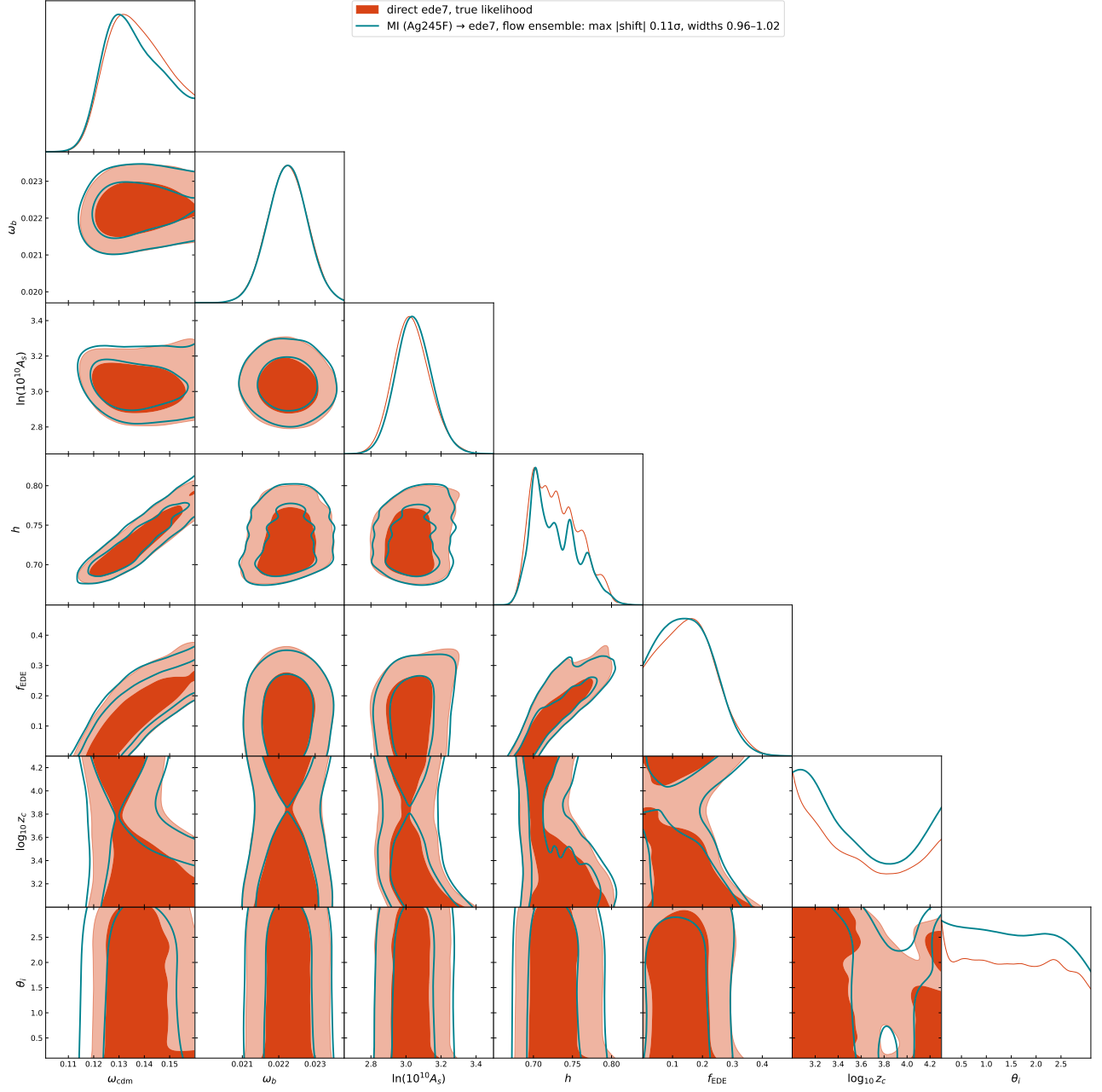


Figure 4: Early dark energy (n_s fixed).

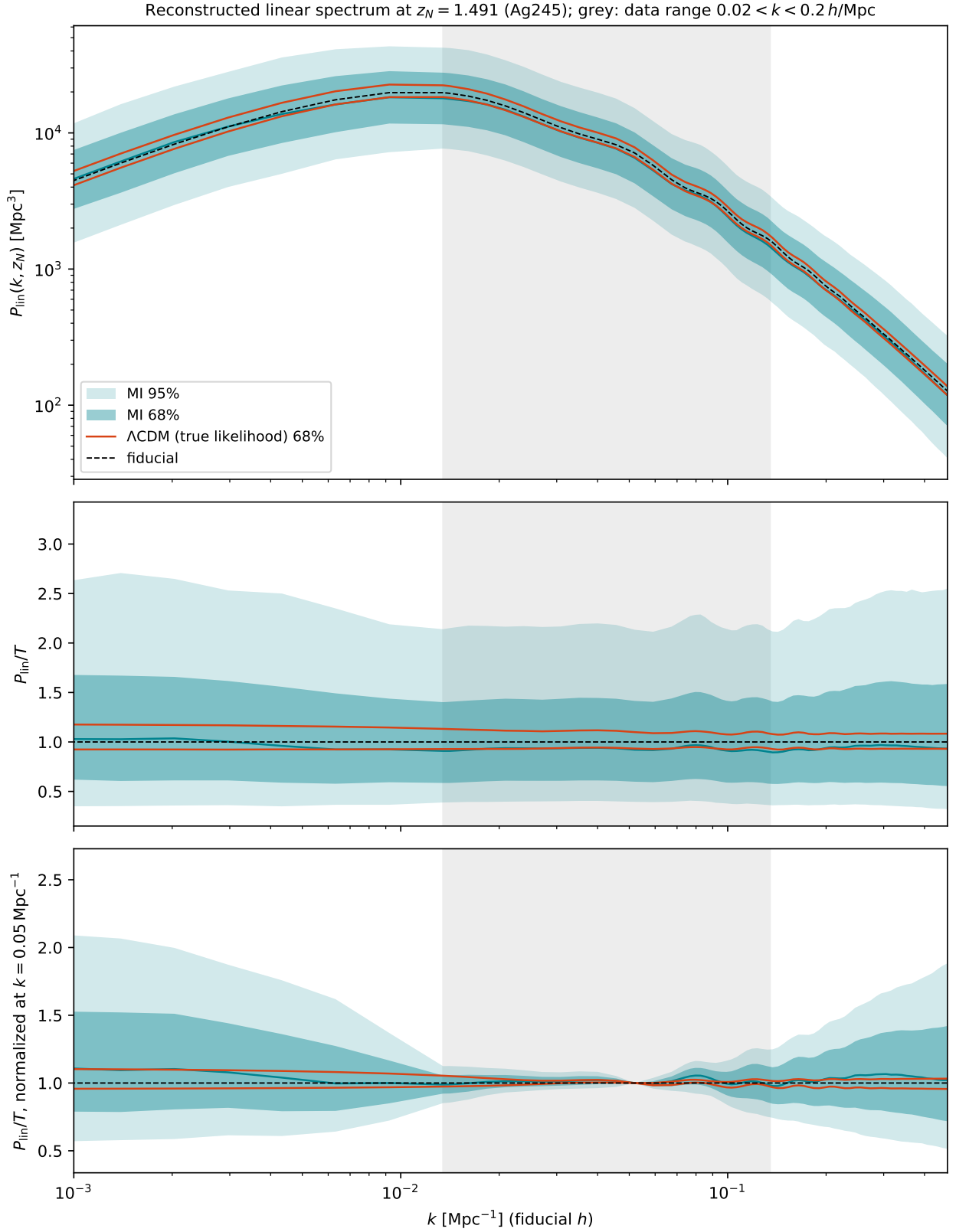


Figure 5: The reconstructed $P_{\text{lin}}(k, z_N = 1.491)$: MI posterior (68% and 95%) against the ΛCDM true-likelihood posterior (red, 68%). Top: absolute; middle: ratio to the fiducial; bottom: shape (ratio normalized at $k = 0.05 \text{ Mpc}^{-1}$). Grey: the data range.

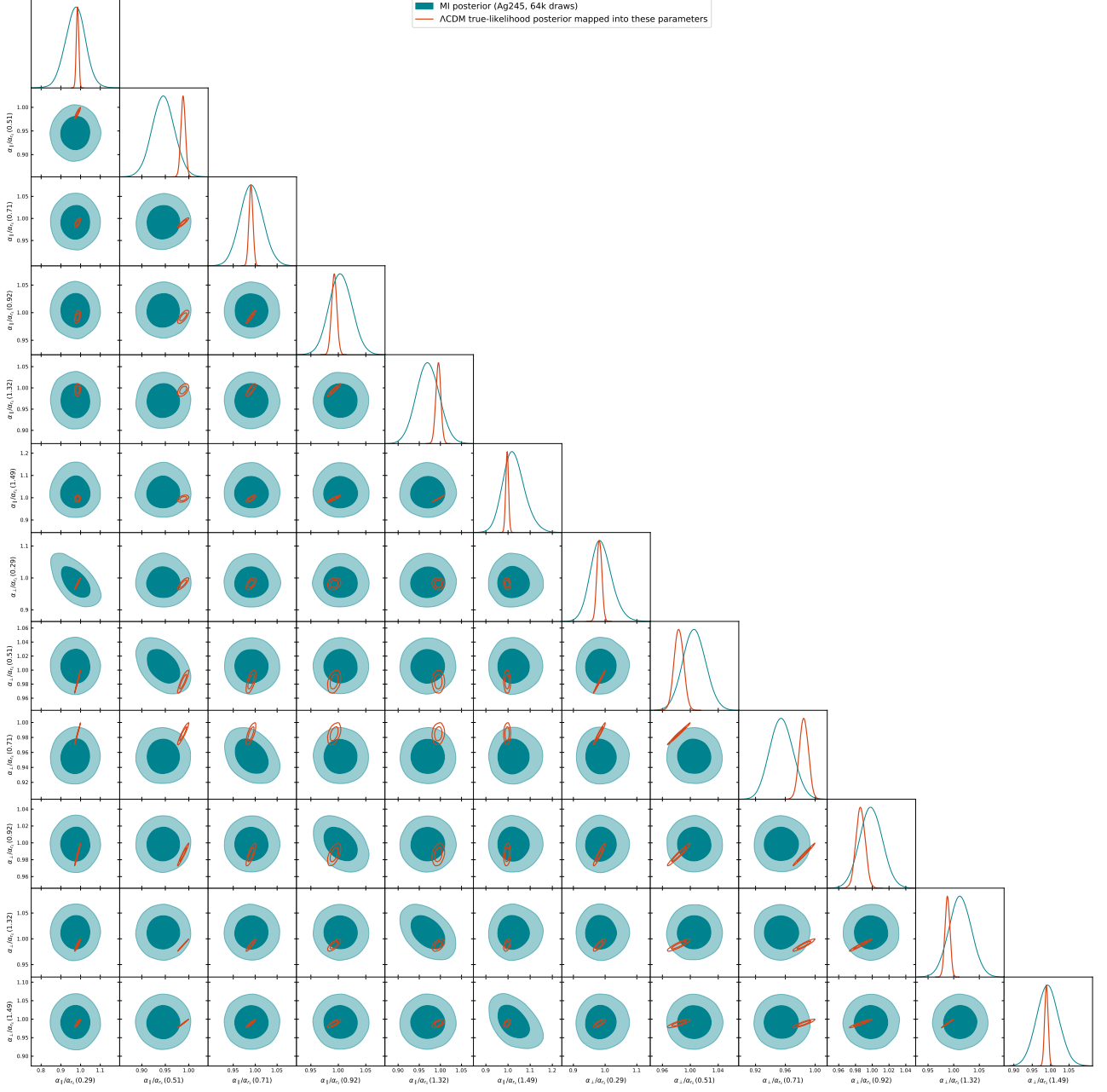


Figure 6: MI posterior of the BAO-equivalent dilations $\alpha/\alpha_{r_s} = (D/r_d)/(D/r_d)_{\text{fid}}$ at the six redshifts, with the Λ CDM posterior mapped in.

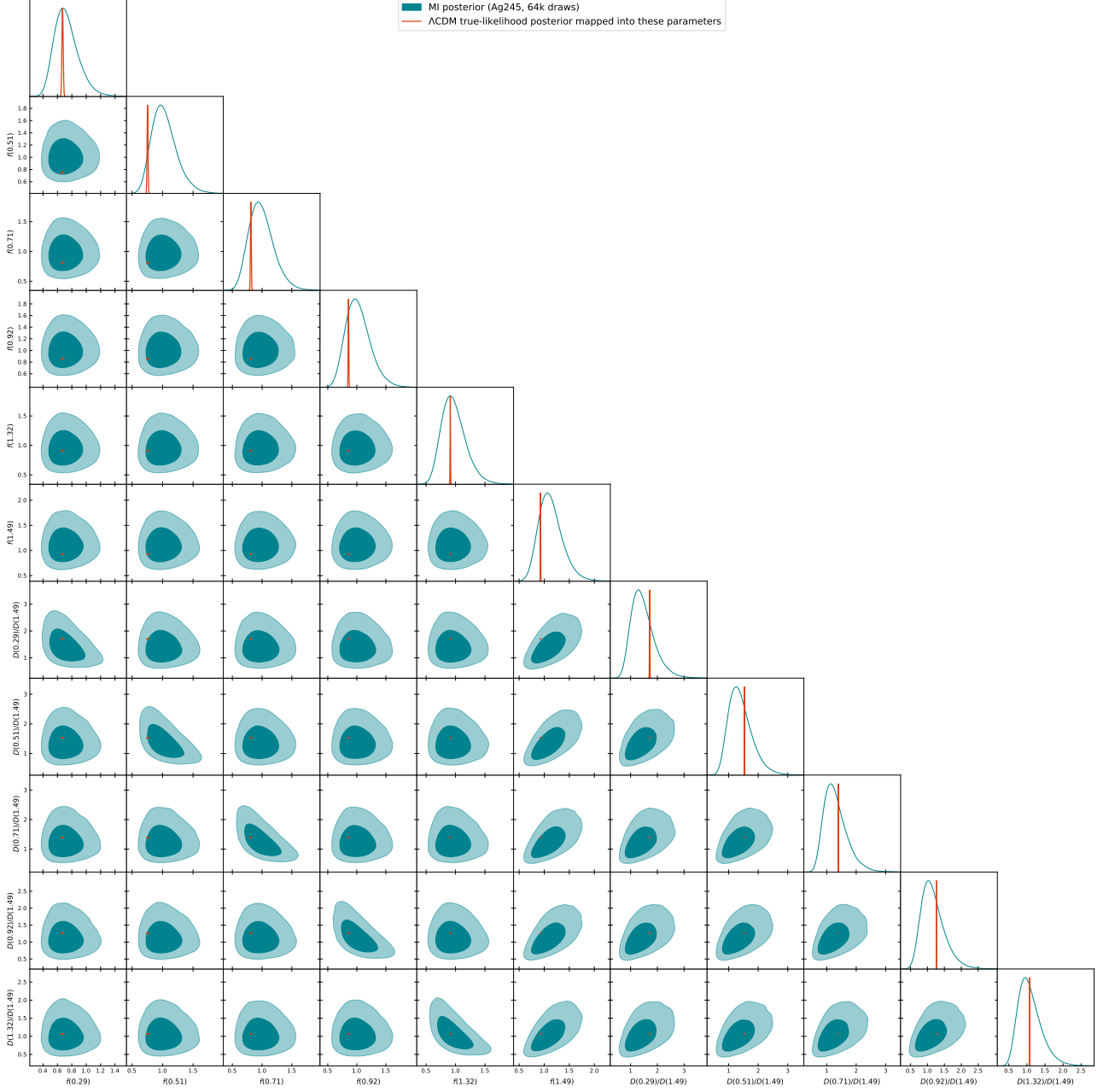


Figure 7: MI posterior of the growth rates $f(z)$ and the growth ratios $D(z)/D(z_N)$, with Λ CDM mapped in.

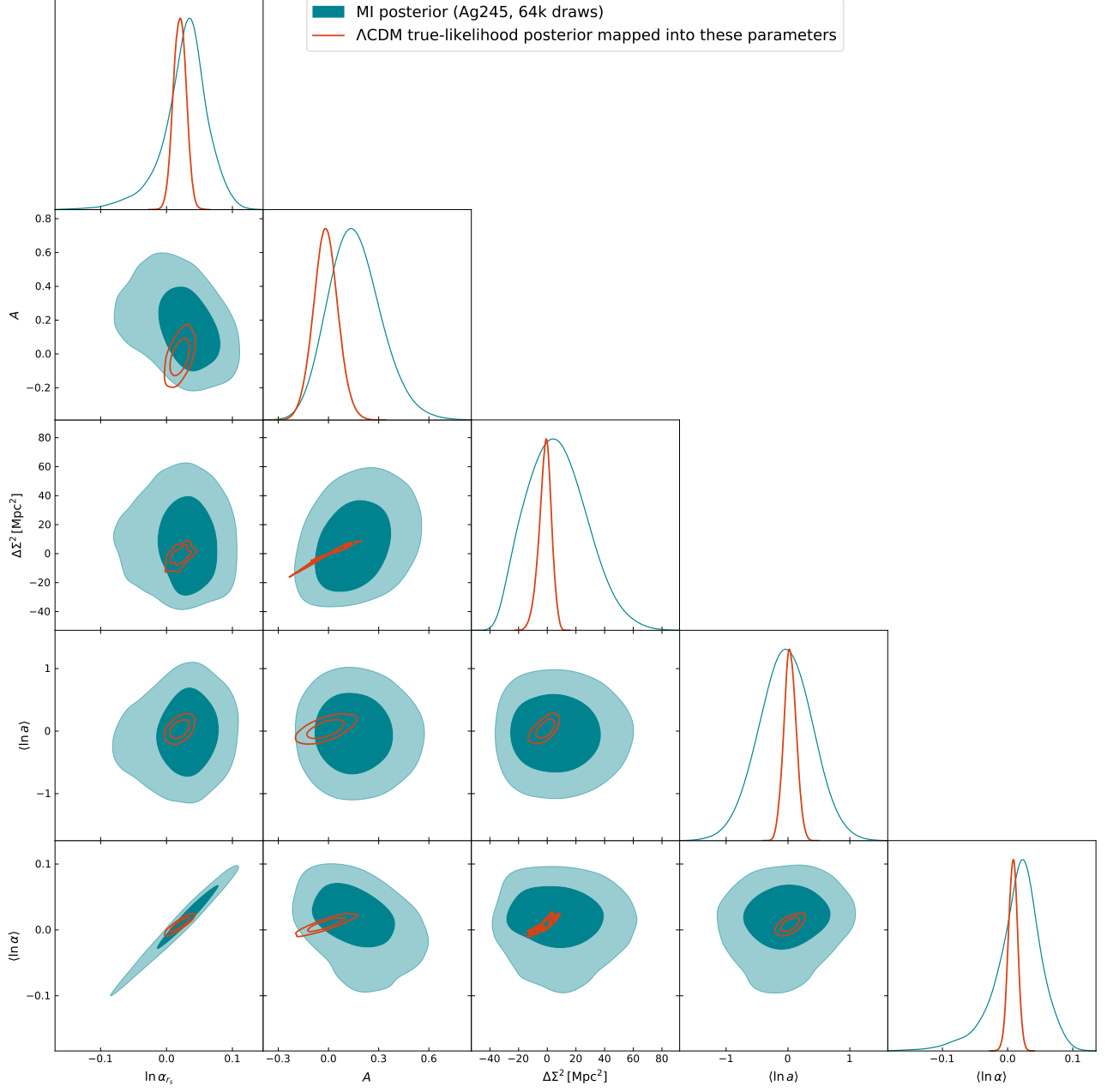


Figure 8: MI posterior of α_{r_s} , A , $\Delta \Sigma^2$, the amplitude at z_N $\langle \ln a \rangle$ and the mean dilation $\langle \ln \alpha \rangle$, with Λ CDM mapped in (red).